

10.18 Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

- a. Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?

```
> cat(sprintf("母親有大學學歷 (MOTHERCOLL=1) 的比例: %.2f%%\n",
+             mean(mroz1$MOTHERCOLL) * 100))
母親有大學學歷 (MOTHERCOLL=1) 的比例: 12.15%
> cat(sprintf("父親有大學學歷 (FATHERCOLL=1) 的比例: %.2f%%\n",
+             mean(mroz1$FATHERCOLL) * 100))
父親有大學學歷 (FATHERCOLL=1) 的比例: 11.68%
> cat(sprintf("父母至少一方有大學學歷的比例: %.2f%%\n",
+             mean(mroz1$MOTHERCOLL == 1 | mroz1$FATHERCOLL == 1) * 100))
父母至少一方有大學學歷的比例: 18.69%
```

- b. Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?

相關係數矩陣:

```
> print(round(cor_matrix, 4))
```

	educ	MOTHERCOLL	FATHERCOLL	mothereduc	fathereduc
educ	1.0000	0.3595	0.3985	0.3870	0.4154
MOTHERCOLL	0.3595	1.0000	0.3546	0.6128	0.3365
FATHERCOLL	0.3985	0.3546	1.0000	0.3438	0.6709
mothereduc	0.3870	0.6128	0.3438	1.0000	0.5541
fathereduc	0.4154	0.3365	0.6709	0.5541	1.0000

MOTHERCOLL/*FATHERCOLL* 為二元虛擬變數，只捕捉「是否受過大學教育」的門檻效果，可避免原始教育年數在高值時的衡量誤差，因此可能是更乾淨的工具變數。

- c. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?

```
Call:
lm(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
    I(exper^2) + MOTHERCOLL, data = mroz1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.08719	-0.32444	0.04147	0.36634	2.35621

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.1327561	0.4965325	-0.267	0.78932
educ	0.0760180	0.0394077	1.929	0.05440 .
exper	0.0433444	0.0134135	3.231	0.00133 **
I(exper^2)	-0.0008711	0.0004017	-2.169	0.03066 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

EDUC 係數的 95% 信賴區間:

```
> print(ci_c["educ", ])
2.5 % 97.5 %
-0.001219763 0.153255678
```

EDUC 係數 ($\beta_2 = 0.0760$): 在控制工作經驗後，教育年數每增加 1 年，預期工資約上升 7.60%。然而 p 值為 0.054，僅在 10% 顯著水準下邊緣顯著 (未達 5%)。EDUC 的 95% 信賴區間為 (-0.0012, 0.1533): 區間涵蓋 0，代表在 5% 顯著水準下，無法拒絕 EDUC 對 $\ln(\text{WAGE})$ 無影響的虛無假說。EXPER 與 EXPER^2 : 工作經驗對薪資有顯著的非線性正向影響 (EXPER 在 1% 顯著， EXPER^2 在 5% 顯著)，符合一般人力資本理論預期。

以 *MOTHERCOLL* 為工具變數的 IV 估計顯示，教育對薪資的報酬約為每年 7.6%，但由於單一工具變數的估計精確度有限，標準誤較大 (0.0394)，導致信賴區間較寬且涵蓋 0。因此在 5% 顯著水準下，教育報酬的統計顯著性不足。

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-Squared: 0.147, Adjusted R-squared: 0.1409
Wald test: 8.2 on 3 and 424 DF, p-value: 2.569e-05

- d. For the problem in part (c), estimate the first-stage equation. What is the value of the *F*-test statistic for the hypothesis that *MOTHERCOLL* has no effect on *EDUC*? Is *MOTHERCOLL* a strong instrument?

```
Call:
lm(formula = educ ~ exper + I(exper^2) + MOTHERCOLL, data = mroz1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-7.4267	-0.4826	-0.3731	1.0000	4.9353

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.079094	0.303118	39.849	< 2e-16 ***
exper	0.056230	0.042101	1.336	0.182
I(exper^2)	-0.001956	0.001256	-1.557	0.120
MOTHERCOLL	2.517068	0.315713	7.973	1.46e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.133 on 424 degrees of freedom
Multiple R-squared: 0.1347, Adjusted R-squared: 0.1285
F-statistic: 21.99 on 3 and 424 DF, p-value: 2.965e-13

F 檢定 (H_0 : *MOTHERCOLL* 對 *EDUC* 無影響):

```
> print(f_test_d)
```

Linear hypothesis test:
MOTHERCOLL = 0

Model 1: restricted model

Model 2: *educ* ~ *exper* + *I(exper^2)* + *MOTHERCOLL*

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	424	1929.9	1	289.32	63.563	1.455e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

MOTHERCOLL 係數 (2.517): 母親有大學學歷的女性，其本身教育年數平均比母親無大學學歷者多約 2.52 年，且在 1% 顯著水準下高度顯著 ($p < 2e-16$)。*MOTHERCOLL* 與 *EDUC* 的相關性強且顯著， $F = 63.56 \gg 10$ ，依據 Staiger & Stock (1997) 的經驗法則，*MOTHERCOLL* 為強工具變數 (strong instrument)，不存在弱工具變數問題，(c) 小題的 IV 估計是可信的。

e. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
      I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz1)

Residuals:
    Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819   0.3922213  -0.712   0.47714
educ         0.0878477   0.0307808   2.854   0.00453 **
exper        0.0426761   0.0132950   3.210   0.00143 **
I(exper^2)   -0.0008486   0.0003976  -2.135   0.03337 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153, Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06
```

```
EDUC 係數的 95% 信賴區間:
> print(ci_e["educ", ])
      2.5 %      97.5 %
0.02751845 0.14817686
```

(c) 信賴區間寬度 = 0.1545, (e) 信賴區間寬度 = 0.1207
(e) 的信賴區間較窄，兩個工具變數提升了估計精確度。

加入第二個工具變數 *FATHERCOLL* 後，額外的工具變數資訊使估計標準誤從 0.0394 降至 0.0308，信賴區間明顯變窄，*EDUC* 的統計顯著性也從邊緣顯著提升至 1% 顯著水準。兩個工具變數的過度識別設定 (over-identified) 在估計效率上優於 (c) 的剛好識別 (just-identified)

f. For the problem in part (e), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?

```
Call:
lm(formula = educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL,
    data = mroz1)

Residuals:
    Min       1Q   Median       3Q      Max
-7.2152 -0.3056 -0.2152  0.7627  5.0620

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 11.890259   0.290251  40.965 < 2e-16 ***
exper        0.049149   0.040133   1.225   0.221
I(exper^2)   -0.001449   0.001199  -1.209   0.227
MOTHERCOLL   1.749947   0.322347   5.429 9.58e-08 ***
FATHERCOLL   2.186612   0.329917   6.628 1.04e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.033 on 423 degrees of freedom
Multiple R-squared: 0.2161, Adjusted R-squared: 0.2086
F-statistic: 29.15 on 4 and 423 DF, p-value: < 2.2e-16
```

```
聯合 F 檢定 (H0: MOTHERCOLL 與 FATHERCOLL 同時對 EDUC 無影響):
> print(f_test_f)
```

Linear hypothesis test:
MOTHERCOLL = 0
FATHERCOLL = 0

Model 1: restricted model
Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	423	1748.3	2	470.88	56.963	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

F > 10，工具變數組合足夠強。

g. For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

```
Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
      I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz1)

Residuals:
    Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819   0.3922213  -0.712   0.47714
educ         0.0878477   0.0307808   2.854   0.00453 **
exper        0.0426761   0.0132950   3.210   0.00143 **
I(exper^2)   -0.0008486   0.0003976  -2.135   0.03337 *

Diagnostic tests:
      df1 df2 statistic p-value
Weak instruments  2  423    56.963 <2e-16 ***
Wu-Hausman       1  423     0.519   0.472
Sargan           1   NA     0.238   0.626
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153, Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06
```

[Sargan 檢定解讀]

```
> cat("H0: 多餘工具變數 (FATHERCOLL) 與誤差項無相關 (工具變數外生有效) \n")
H0: 多餘工具變數 (FATHERCOLL) 與誤差項無相關 (工具變數外生有效)
> cat("p-value > 0.05 → 不拒絕 H0, FATHERCOLL 通過外生性檢定, 工具變數設定有效.\n")
p-value > 0.05 → 不拒絕 H0, FATHERCOLL 通過外生性檢定, 工具變數設定有效。
> cat("p-value < 0.05 → 拒絕 H0, 工具變數可能與誤差項相關, 設定有問題.\n")
p-value < 0.05 → 拒絕 H0, 工具變數可能與誤差項相關, 設定有問題。
```


10.20 The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the j th security's "beta" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

- a. Use the observations on Microsoft in the data file `capm5` to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?

```
Call:
lm(formula = excess_msft ~ excess_mkt, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27424 -0.04744 -0.00820  0.03869  0.35801

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003250   0.006036   0.538   0.591
excess_mkt   1.201840   0.122152   9.839 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523,    Adjusted R-squared:  0.3486
F-statistic: 96.8 on 1 and 178 DF, p-value: < 2.2e-16
```

OLS 估計 $\beta = 1.2018$ ，在 1% 顯著水準下高度顯著 ($p < 2e-16$)。市場超額報酬每上升 1%，Microsoft 超額報酬約上升 1.20%。截距 $\alpha = 0.003250$ 不顯著 ($p = 0.591$)，與 CAPM 理論 ($\alpha = 0$) 一致。
結論： $\beta > 1$ ，Microsoft 屬於高風險積極型股票，系統性風險高於市場平均。

- b. It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \dots, 180$. Does this variable potentially satisfy the conditions IV1–IV3? Create $RANK$ and obtain the first-stage regression results. Is the coefficient of $RANK$ very significant? What is the R^2 of the first-stage regression? Can $RANK$ be regarded as a strong IV?

```
RANK 變數摘要:
> print(summary(capm5$RANK))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.00  45.75   90.50   90.50  135.25   180.00

Call:
lm(formula = excess_mkt ~ RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.110497 -0.006308  0.001497  0.009433  0.029513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.903e-02  2.195e-03  -36.0 <2e-16 ***
RANK         9.067e-04  2.104e-05   43.1 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126,    Adjusted R-squared:  0.9121
F-statistic: 1858 on 1 and 178 DF, p-value: < 2.2e-16
```

$RANK$ 係數 = 0.0009067， $t = 43.1$ ，高度顯著。第一階段 $R^2 = 0.9126$ ， $F = 1857.59$ ，遠超門檻值 10。
 $RANK$ 為純統計排序變數，不直接影響 Microsoft 報酬，外生性合理，滿足 IV1–IV3 條件。
結論： $RANK$ 為強工具變數，可有效用於 2SLS 估計。

- c. Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?

```
Call:
lm(formula = excess_msft ~ excess_mkt + v_hat_c, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.005984   0.504   0.6146
excess_mkt   1.278318   0.126749  10.085 <2e-16 ***
v_hat_c      -0.874599   0.428626  -2.040   0.0428 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,    Adjusted R-squared:  0.36
F-statistic: 51.34 on 2 and 177 DF, p-value: < 2.2e-16
```

第一階段殘差 $\hat{v}$ 加入擴增迴歸後， $\hat{v}$ 的係數 = -0.8746 ， p 值 = 0.0428，但未達 1% 顯著水準。
結論：在 1% 顯著水準下不拒絕外生性，無法確認市場超額報酬具有內生性。 p 值接近 5% 邊界，測量誤差問題仍值得留意。

d. Use RANK as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

```
Call:
ivreg(formula = excess_msft ~ excess_mkt | RANK, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.271625 -0.049675 -0.009693  0.037683  0.355579

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.006044   0.499   0.618
excess_mkt   1.278318   0.128011   9.986  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-squared:  0.3508,    Adjusted R-squared:  0.3472
Wald test: 99.72 on 1 and 178 DF, p-value: < 2.2e-16
```

IV 估計 $\beta = 1.2783$ ，高於 OLS (1.2018)，差距 +0.0765 (+6.4%)。
結論：IV 估計高於 OLS，符合衰減偏誤理論預期——OLS 因測量誤差低估 β ，IV 修正後上升。
兩者差距不大，暗示測量誤差程度相對有限。

e. Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both RANK and POS as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?

```
Call:
lm(formula = excess_mkt ~ RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.109182 -0.006732  0.002858  0.008936  0.026652

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0804216   0.0022622  -35.55  <2e-16 ***
RANK         0.0009819   0.0000400   24.55  <2e-16 ***
POS        -0.0092762   0.0042156   -2.20   0.0291 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149,    Adjusted R-squared:  0.9139
F-statistic: 951.3 on 2 and 177 DF, p-value: < 2.2e-16
```

```
聯合 F 檢定 (H0: RANK 與 POS 同時對市場超額報酬無影響):
> print(f_test_e)

Linear hypothesis test:
RANK = 0
POS = 0

Model 1: restricted model
Model 2: excess_mkt ~ RANK + POS

   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     179 0.43784
2     177 0.03727  2     0.40057 951.26 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

POS = 1 的比例為 58.33%。加入 POS 後，聯合 F = 951.26， $R^2 = 0.9149$ ，兩個工具變數對市場超額報酬的解釋力極強。
結論：聯合 F >> 10，RANK 與 POS 組合為強工具變數，可進行過度識別 IV 估計。

f. Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?

```
Call:
lm(formula = excess_msft ~ excess_mkt + v_hat_e, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27132 -0.04261 -0.00812  0.03343  0.34867

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.005972   0.503   0.6157
excess_mkt   1.283118   0.126344  10.156  <2e-16 ***
v_hat_e     -0.954918   0.433062  -2.205   0.0287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,    Adjusted R-squared:  0.3625
F-statistic: 51.88 on 2 and 177 DF, p-value: < 2.2e-16
```

使用雙 IV 第一階段殘差， $\hat{v}$ 的 p 值 = 0.0287，在 5% 水準下顯著，但仍未達 1% 顯著水準。
結論：與 (c) 結論一致，在 1% 水準下不拒絕外生性。p 值從 0.0428 降至 0.0287，雙 IV 檢定稍更靈敏，但不足以在 1% 水準下確認內生性。

g. Obtain the IV/2SLS estimates of the CAPM model using *RANK* and *POS* as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

```
Call:
ivreg(formula = excess_msft ~ excess_mkt | RANK + POS, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27168 -0.04960 -0.00983  0.03762  0.35543

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004    0.006044   0.497    0.62
excess_mkt    1.283118    0.127866  10.035 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared: 0.3507,    Adjusted R-squared: 0.347
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16
```

雙 IV 估計 $\beta = 1.2831$ ，與單一 IV (1.2783) 差距僅 0.0048，標準誤微降 (0.1280 $\rightarrow$ 0.1279)。

三種估計排列：OLS (1.2018) < IV-RANK (1.2783) < IV-RANK+POS (1.2831)。

結論：方向一致，符合衰減偏誤修正預期。加入 POS 精確度略提升，但結論與 (d) 相同，Microsoft 為高風險股票。

h. Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

```
> cat(sprintf("樣本數 n = %d\n", n))
樣本數 n = 180
> cat(sprintf("殘差迴歸 R^2 = %.6f\n", r2_sargan))
殘差迴歸 R^2 = 0.003103
> cat(sprintf("Sargan 統計量 (n*R^2) = %.4f\n", sargan_stat))
Sargan 統計量 (n*R^2) = 0.5585
> cat(sprintf("自由度 = %d (多餘工具變數數) \n", df_sargan))
自由度 = 1 (多餘工具變數數)
> cat(sprintf("χ^2 臨界值 (5%, df=1) = %.4f\n", chi_critical))
χ^2 臨界值 (5%, df=1) = 3.8415
> cat(sprintf("p-value = %.4f\n", pval_sargan))
p-value = 0.4549
```

取 (g) 的 2SLS 殘差對 RANK、POS 迴歸， $nR^2 = 180 \times 0.003103 = 0.5585$ ，自由度 = 1 (多餘工具變數 1 個)， χ^2 臨界值 (5%) = 3.8415，p 值 = 0.4549。

結論：0.5585 << 3.8415，不拒絕 H_0 ，POS 通過外生性檢定，與誤差項無顯著相關。搭配聯合 $F = 951.26$ ，RANK 與 POS 同時滿足相關性與外生性條件，工具變數設定有效。